Moon Orbiter for Water Imaging (MOWI)

# Avionics Proposal

Group 1



CONTRIBUTORS: 6969623, 6941657, 6943841

# 1 Executive Summary

"Together with other ESA member states, a nanosatellite is being procured and sent into a lunar orbit for water imaging the poles for a 4-year mission in April 2032" [1].

The objectives for this mission are as follows:

- **OB-1**: Identify and map the distribution of water ice ($H_2O$) at both the lunar poles (60°S–90°S, 60°N–90°N).
- **OB-2**: Maximise the use of COTS products for cost-effectiveness.

60° was chosen because of earlier findings [2] showing very little water elsewhere.

The Moon Orbiter for Water Imaging (MOWI) will be reliant on the Lunar Pathfinder [3] (also referred to as the "hub") for all communications to and from Earth. The expected deployment of MOWI in lunar orbit is shown in Figure 1 below.

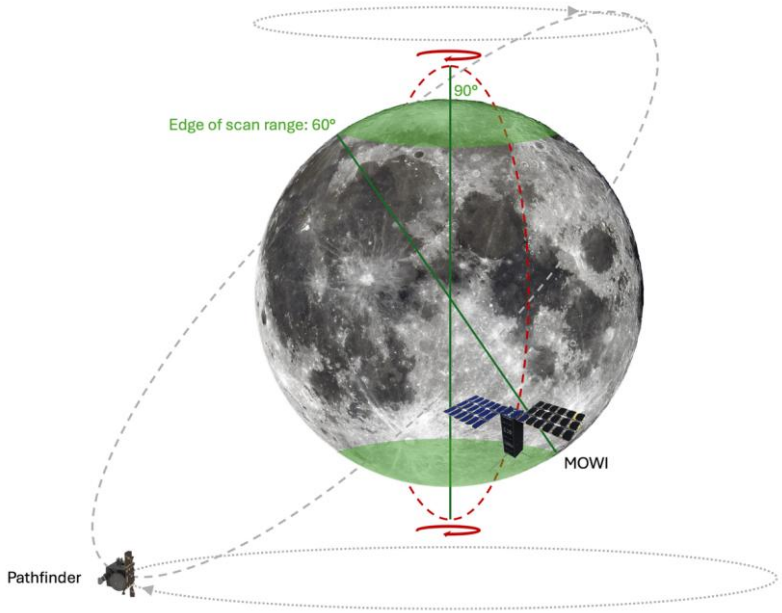



*Figure 1 - MOWI and Pathfinder orbits*

MOWI's output will consist of a topological water density map of both poles constructed and transmitted every ~2-hour orbit over the 4-year mission duration.

This report has concluded that COTS products can be reliably used for all major Power, OBC and Comms sub-systems, accepting some additional risk as some of the components are only at TRL6 [4] for lunar orbit. This conclusion does not extend to Earth-targeted COTS products.

# 2 Architecture and Interfaces

## 2.1 External Interfaces

The only relevant external system during the Lunar Orbit phase is the Lunar Pathfinder, which MOWI will be relying on as a communications relay.

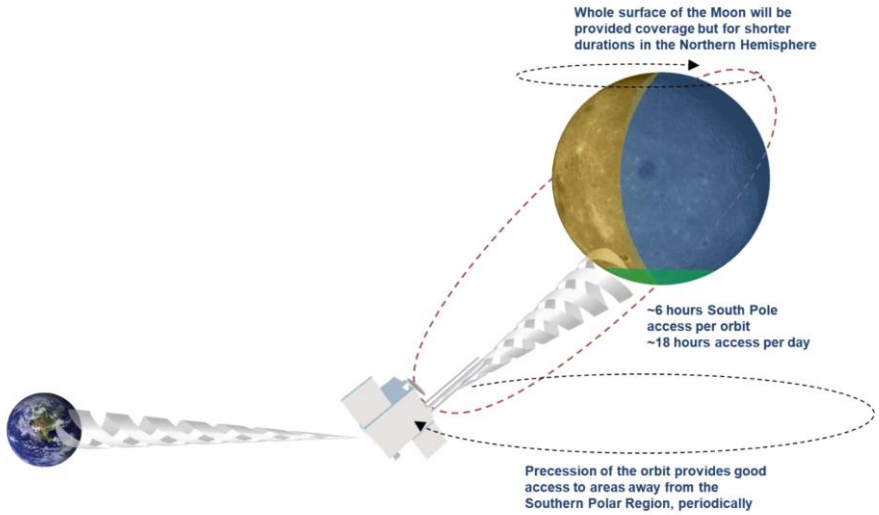



*Figure 2 - Lunar Pathfinder orbit [3]*

We will be using Pathfinder's "Autonomous service" package for managing communications as the most cost-effective option. In this autonomous mode, the frequency, duration and time of the contact between the Lunar Pathfinder spacecraft and MOWI is automatically optimised via algorithm.

## 2.2 Internal Architecture

The main system components and interfaces are outlined in Figure 3 below. The VMMO [5] mission is used as a reference implementation for those selections not covered in more detail elsewhere in this report.

The overall architectural design is centralised, using the "star" design pattern. Given the relatively simple design for this satellite and the fixed overall mission requirements, it is unlikely that additional systems would need to be added later, so the downside of using this pattern with respect to node addition overhead is mitigated.

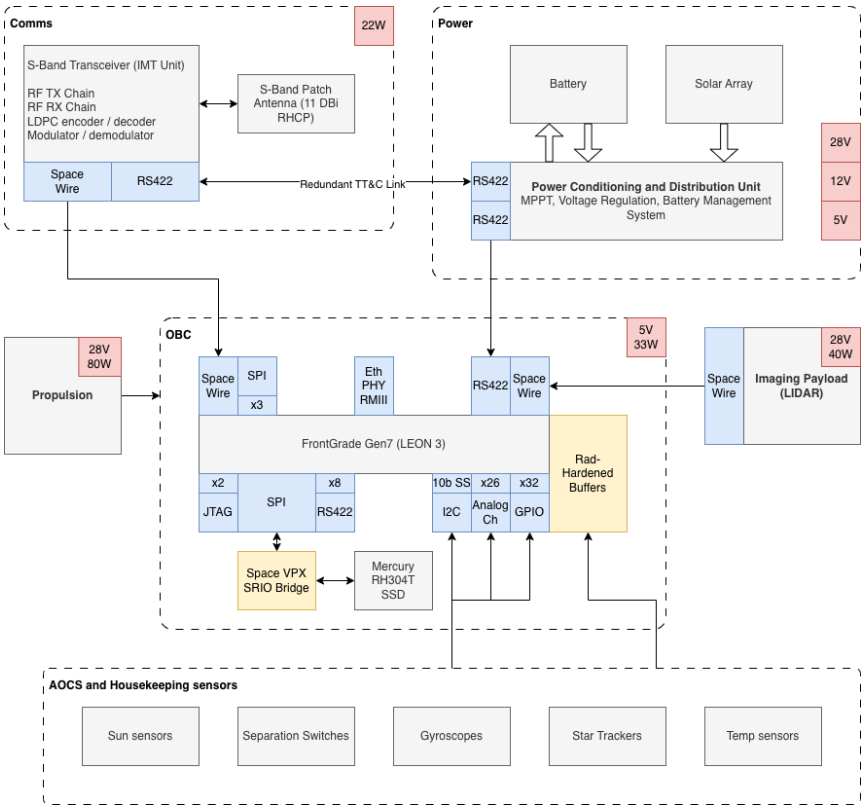



*Figure 3 - MOWI Architecture*

### 2.2.1 Interfaces

Some compromises had to be made for interfacing to enable the best selection of COTS products possible. Not all COTS components had similar interfaces so, for example, a bespoke SpaceVPX SRIO bridge will need to be included to interface the SSD with the OBC.

Sensors across the satellite will likewise require their own interfaces. Analog sensors can be handled directly on the OBC via the analog channels provided, however more complex sensors such as Star Trackers and Gyroscopes will likely require more specialist interfaces that may not exist out of the box on the OBC. These will likewise therefore require bespoke interfacing logic.

### 2.2.2 Redundancy

Some redundancy has been designed into the system, such as the redundant TT&C link direct to the power sub-system from the comms sub-system to enable direct control over power-cycling of all components including the OBC. In the event of unexpected failures that could, for example, be caused by radiation induced latchup events that could be resolved by power

cycling, this redundant link can be used to regain control of the satellite when it would otherwise be lost without the OBC.

### 2.2.3 Busses

Rather than use a single bus to connect all systems, the centralised “star” pattern has been used in this design. This allows us greater freedom with respect to interfaces available on COTS products and the advantage of flexibility that a bus would normally provide is mute given we have a fixed mission. The downside is that we will need to consider redundancy and resiliency on all connections independently.

# 3 Concept of Operations – Lunar Orbit Phase

## 3.1 Operations Environment

MOWI will be orbiting the Moon in a circular polar orbit at 100km as shown in Figure 1. During its orbit, there will be periods during which MOWI has direct line of sight with the Lunar Pathfinder and it will be open to communication between the two. The orbital parameters for the spacecraft can be found in Appendix A: Orbital Parameters.

The lunar orbital environment can be considered equivalent to interstellar space as the Moon provides no geomagnetic shielding or atmosphere. The primary factors of note are as follows:

- **Radiation** – The radiation environment is dominated by solar protons and Galactic Cosmic Rays (GCRs) [6] and there is no protection from Solar Particle Events (SPEs). Single Event Upsets (SEUs) in the order $10^{-6}$ per bit per day are expected as a result this. In addition, ionising radiation in the order of $5x10^{-7}$ rads per second in Silicon [7] is expected even behind 100mil of Aluminium shielding. Over the course of the mission duration, this will total approximately 63 rads.
- **Temperature range** – +127°C to -183°C [8] imparted by a combination of solar radiation and warm body radiation from the Moon.

## 3.2 Nominal Condition Operations

During its orbital cycle, MOWI will run 3 standard modes of operation. To ensure mission success, the operating modes cannot be mutually exclusive (for example, the comms window may by necessity overlap with the imaging window):

- **Nominal mode** - Only 'housekeeping' data is stored.
- **Payload collection** - Data is routed from the payload to the OBC to storage.
- **Data downlink** - Data is routed from Storage to the OBC and then to the hub.

Activation of these modes is determined primarily by the location of MOWI and by its position relative to Lunar Pathfinder as shown in Figure 4.

We are choosing to use the Pathfinder's "Autonomous" SLA for communication with MOWI and therefore the links will be initiated by Pathfinder.

A likely exemplar pattern for MOWI activating modes during a single orbit is visualised in Figure 5.

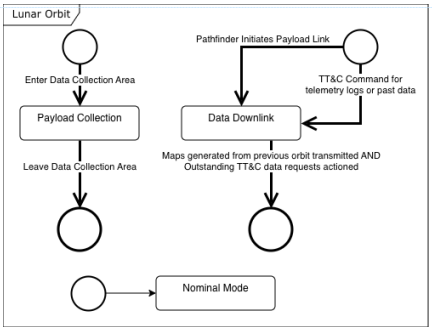



*Figure 4 - MOWI Operational Mode Transition Diagram*

**Commented [CRC(CS&EE1]:** This needs to be changed. The modes cannot be mutually exclusive if we are to meet mission requirements.

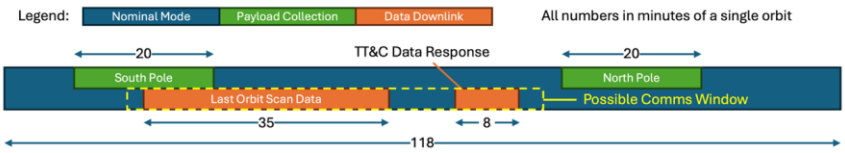



*Figure 5 - A nominal operation conditions exemplar of mode transitions per orbit*

### 3.2.1 Nominal Mode

During normal operations, Nominal Mode is active. MOWI collects housekeeping data across all sensors and performs station-keeping adjustments via propulsion systems and AOCS as required.

### 3.2.2 Data Downlink Mode

This mode is triggered by Lunar Pathfinder initiating links to MOWI during line of sight between the two spacecraft. During data downlink, MOWI will:

- **TT&C Link**: Receive commands and past telemetry or water map data requests and transmit a stream of current telemetry.
- **Payload Link**: Transmit the last orbit's worth of map data and any additional historic data requested via commands received on the TT&C Link.

At the closest separation between the two, MOWI can transmit payload data at the maximum 1 Mbps data rate for a period of about 20 minutes. This drops to 256kbps and 48.8 kbps for the rest of the 35-minute contact window as the orbital separation increases.

**Service Option: Autonomous Mode**
In the "autonomous" mode, an agreed amount of data per day is relayed between earth ground station and relay satellite. This is called "autonomous" as Lunar Pathfinder optimises automatically when the customer asset is contacted for data transfer, within an agreed periodicity. This allows Lunar Pathfinder to automatically offer the best possible data-rates to customers whenever available. [3]

### 3.2.3 Payload Collection Mode

This mode is activated when MOWI reaches ±60° latitude and deactivates again on reaching the same latitude again on the other side of the pole.

In this mode, the onboard LIDAR imaging payload is active, and data captured is routed to storage by the OBC.

## 3.3 Off-Nominal Conditions

Off-nominal conditions are outside the scope of this report and will require further analysis.

# 4 Sub-System Reviews

## 4.1 Communications Subsystem

### 4.1.1 Objectives

The communication subsystem is designed to meet the requirements of MOWI which are to perform Communication in two modes: 48 kbps BPSK specifically for TT&C and 1 Mbps S-band QPSK for Payload downlink, both full duplex, to the Lunar Pathfinder which will relay data back to Earth [1]. Using the Service User Guide of Lunar Pathfinder, an example of User Case #4 – CubeSat has been considered to design the geometry of MOWI [3]. For this mission, as already mentioned in the Executive Summary, a Circular Lower Lunar Orbit (LLO) has been chosen with 100 km altitude above the lunar surface, and the orbital parameters are mentioned in the Appendix A for the calculations. The orbital complete link budget was derived considering the geometry of lunar orbit with minimum separation distance between the MOWI and Lunar Pathfinder to be 600 km and maximum separation distance to be 8000 km due to the elliptical orbit of Lunar Pathfinder for which the calculation will be mentioned in the Appendix __. The avionics platform uses commercial-off-the-shelf technologies with a view to balance the trade-off between functionality and reliability. The variation in the geometry and the COTS chosen drives the communication design, hence, the communication subsystem must:

a. Select COTS with good Antenna gain, Transmitter RF output power, and reasonable Losses to define the Link Budget and enhance the RF performance.
b. Provide a positive link margin to maximise the use of 1 Mbps S-band QPSK at mid-range distance to the Lunar Pathfinder.
c. Maintain TT&C capability at maximum separation.
d. Use LDPC rate = ½ FEC to make extremely efficient use of SNR.
e. Operate within the power subsystem limits.

### 4.1.2 Link Budget Derivation

The Link Budget is defined partially by the selection of COTS equipment and partially by the geometry of the mission since it will provide the values to decide the amount data speed and data that can be send over to the Lunar Pathfinder during the contact period. Now, the S-band transceiver chosen to do the task in an efficient manner is an ‘IMT SBTRcvr S-Band Transceiver’ [9] and the antenna chosen is ‘IQ Spacecom S-band Antenna’ [10]keeping in my mind that it is a COTS equipment. A detailed yet concise review about these choices and their specifications have been explained in the ‘1 COTS vs 1 SpaceTech’ section to reason the calculations of the subsystem. Given the notably lower TT&C uplink rate, 48.8 kbps, the reverse link margin exceeds that of the forward link and does not constraint operations.

#### 4.1.2.1 Data Volume

According to the requirements of the mission, the storage of around 8x6.48 GB per orbit has been asked to be maintained. Therefore, to make this mission work with the mentioned requirements, a solution to the problem of data generation has been given in the OBC

Subsystem section further in the report. From the Communication point of view, calculation needs to be done to provide OBC Subsystem with the maximum usable data per pass.

#### 4.1.2.2 Mission Geometry

It is important to define a mission geometry first before diving into the calculations since it will decide the communication altogether. Taking reference of the orbital parameters from the Lunar Pathfinder CubeSat example [3], the orbital period, $T_{MOWI}$ ~ 117-118 Earth minutes ~ 1.96 Earth Hours, that gives the Number of Orbits ~ 12.2 Orbits per Earth-day. The orbital period of LP from the service manual is $T_{LP}$ ~ 10.8 Earth-hours. The minimum and maximum separation between LP and MOWI and contact time per orbit depending on the Periselene and Aposelene is given in the table below, for further calculation refer to Appendix __ .

| Location | Separation | Contact time per Orbit (Earth-min) |
|---|---|---|
| LP_Periselene | $d_{min}$ ~ 600 km | ~ 30 minutes |
| LP_Aposelene | $d_{max}$ ~ 8000 km | ~ 50 minutes |

#### 4.1.2.3 Downlink Strategy

The number of orbits per earth-day has turned out to be 12.2 but due to difference in the orbits of LP and MOWI the geometrical constraints will arise that will not allow MOWI to be visibility of LP for all ~12 orbits. Hence, realistic assumptions need to be made to estimate the number of favourable orbits which will decide daily data throughput.

**Assumptions and given values:** [11]

a. Frequency 2.25 GHz (S-band) [12] : The mission specified S-band operation.
b. TX RF power ~ 5 W [9]: The transceiver datasheet mentions this power.
c. AntennaGain~11dBi [10]:The antenna datasheet mentions this gain.
d. Cable Losses~ 1 dB [13]: Any typical short coax at S-band lies between 0.5-1.5 dB.
e. Other system losses ~ 2 dB [13]: These include polarisation mismatch, implementation losses, pointing error loss, connector mismatch etc.
f. Coding=LDPC rate ½ [14]
g. Protocol Overhead=10% [13]: Typical overhead protocol ranges from 10-20%
h. G/T = -10.13 dB/K [3]:Taken directly from Lunar Pathfinder service documentation.
i. 3 favourable passes per Earth-day: Based on relative orbital geometry, solution to which is given in Appendix E.
j. ~40-50 minutes visibility per pass: Based on relative orbital geometry, solution to which is given in Appendix E.
k. Separation varies 600-8000 km during pass: Directly from Pathfinder orbital parameters. Based on relative orbital geometry, solution to which is given in Appendix E.
l. Adaptive Rate Switching: Link Margin decreases with distance.
   - 1 Mbps ~ 20 mins
   - 256 kbps ~ 12 mins
   - 48.8 kbps ~ 8 mins
m. Polarisation~ RHCP: Preferred
n. PA efficiency = 35% = 0.35

Considering these input parameters, the following result has been derived:

| Separation | Duration (Earth-minutes) | Mode | Data per pass | Usable data per favourable orbit pass | **Total usable data for favourable contact orbits** |
|---|---|---|---|---|---|
| Close Approach | 20 | 1 Mbps | 67.5 MB | ~ 79.178 MB | **~178.1-237.5 MB** |
| Mid-range approach | 12 | 256 kbps | 10.368 MB | | |
| Far range approach | 8 | 48.8 kbps | 1.31 MB | | |

**Electrical Power Consumption:**

RF output = 5 W
Baseband + Modulator + Synthesizer = 6-8 W ~ 7 W
Total TX electrical = 22 W

| Mode | RF Power | Electrical |
|---|---|---|
| 1 Mbps | 5 W | 22 W |
| 256 kbps | 3 W | 16-18 W |
| 48.8 kbps | 1 W | 12-14 W |

In conclusion, at 600 km and 3000 km separation, the link margin gives positive values, hence, confirming strong performance near Pathfinder periselene validating the choice of 5 W RF and 11 dBi antenna gain. However, at maximum separation, the 1 Mbps link falls below threshold and gives a negative value confirming that continuous high-rate transmission is not feasible across the full pass. The adoption of LDPC coding at ½ rate ensures a required Eb/No of approximately 2 dB which improves the link robustness in lunar radiation environment. The effective payload rate at 1 Mbps is approximate 450 kbps accounting for the coding and protocol overhead of 10%. Under the assumed moderate visibility scenario of 3/12 favourable orbits of 30-50 minutes pass per Earth Day, the achievable daily downlink capacity approximately is 178.1-237.5 MB per Earth-day.

### 4.1.3 1 COTS vs 1 SpaceTech

#### 4.1.3.1 Transceiver

| **Parameter** | **COTS - IMT SBTRcvr** | **SpaceTech - Rakon NewSpace** |
|---|---|---|
| Frequency | 2200-2300 MHz | 2.2-2.29 GHz |
| Max RF Output | 1-5 W | 2 W |
| Max Data Rate | 1 Mbps (10 Mbps optional) | 5 Mbps |
| Modulation | BPSK/QPSK | BPSK/QPSK/GMSK |
| Full Duplex | Yes | Yes |
| Coding Support | BPSK/QPSK/LDPC | RS, Convolutional, LDPC |
| Peak TX Power Consumption | ~12 W (1 W RF) | <12 W on TX |
| Mass | < 1 kg | 300 g |
| Radiation TID | <20k rad | 50k rad |
| SEE Protection | Limited | No SEL up to LET=43 |
| Lifetime | 3 years | 1-12 years |

**Performance Implications: Justifications**

| Transceiver | Advantages | Limitations |
|---|---|---|
| 1. IMT SBTRcvr | • Enables ~17 dBW EIRP with 11 dBi Antenna. Thus IMT satisfies margin at mid-range.<br>• Radiation hardness considered at 20k rad<br>• The EPS design supports 22 W, hence, IMT remains within acceptable bounds.<br>• Suitable for adaptive rate strategy<br>• More than 3 years of lifetime | • Lower radiation qualification.<br>• No explicit SEL guarantee. |
| 2. Rakon NewSpace | • Radiation Hardened<br>• SEL protection specified<br>• Supports >= 1Mbps data rate<br>• Longer lifetime | • Maximum 2W RF (insufficient EIRP for 1 Mbps mid-range)<br>• Require higjer gain antenna |

#### 4.1.3.2 Antenna

| Parameter | COTS-IQ Spacecom S-Band Patch | SpaceTech-MMA T-DaHGR Deployable Reflectarray |
|---|---|---|
| Type | Body-mounted | Deployable Reflect array |
| Gain | 6-11 dBi | Very High |
| Frequency | 1.98-2.5 GHz | Wideband Scalable |
| Polarisation | RHCP | Configurable |
| Complexity | Low | High |

**Performance Implications: Justifications**

| Antenna | Advantages | Limitations |
|---|---|---|
| 1. IQ Spacecom Patch | • Meets derived gain requirement(<=11dBi)<br>• Requires no deployment mechanism<br>• Less mass<br>• Compatible with small satellites<br>• Mentions suitability with LEO missions<br>• TRL 9 certified | • Not specifically suitable for deep-space radiation<br>• Lower aperture which leads to limited scalability. |
| 2. MMA T-DaHGR | • High bandwidth<br>• Large aperture hence very high gain<br>• Suitable for SAR/high data rate missions | • Deployment complexity hence hinge failure risk and lower structural uncertainty<br>• Higher mass<br>• Increased ADCS burden<br>• Overdesigned for 1 Mbps S-band link<br>• Higher cost |

## 4.2 Power Subsystem

The design of the Electrical Power System (EPS) is driven by the peak power demands of the imaging payload, Comms, OBC and the requirement for electric propulsion availability during all modes of operations. In the low LLO orbit, MOWI experiences frequent eclipse and charge/discharge cycles, ~18000 cycles (assuming one eclipse per orbit) over the 4-year mission duration. These operations impose specific requirements on the battery capacity, cycle life, solar array sizing, and end of life performance. Table 1 summarizes the subsystem power draws that define the EPS baseline.

| **Spacecraft Systems** | **Power Consumption** |
|---|---|
| Electric Propulsion (EP) | 80W |
| LIDAR Payload (Pyl) | 40W |
| Communication system (Comms) | 22W |
| On Board Computer (OBC) | 60W |

*Table 1: Note that these only describe subsystem level power draw and may not include all components.*

### 4.2.1 Mode-based energy budget

As described in (Nominal Condition Operations), the spacecraft's Electrical Power Subsystem (EPS) operates according to the needs of the 3 modes of operation. Table 2 lists all subsystems that are active during each mode and their respective power draw per orbit.

| Mode | Loads | Total Power (W) | Duty per orbit (mins) | Energy per orbit (Wh) |
|---|---|---|---|---|
| Nominal | EP + OBC +Comms | 162 | ~38 | 102.6 |
| Data Downlink | EP + OBC +Comms | 162 | ~40 | 108 |
| Payload Collection | EP + OBC + Comms + Pyl | 202 | ~40 | 134.7 |

*Table 2: Note that the duty per orbit is derived from Conops based estimates. Nominal mode can work in conjunction with other modes.*

Note that the peak power load will be experienced when in eclipse, and all the systems are running. This represents the worst-case scenario and drives the requirement for battery sizing.

### 4.2.2 Battery sizing

Payload collection may occur in either sunlight or eclipse conditions. Therefore, it is necessary to design the EPS and battery system to handle maximum simultaneous power load experienced when all systems are active. Note, the power for the heater is assumed to be 5W.

$$P_{peak} = 80W + 40W + 22W + 60W + 5W(heater) = 207W$$

Using a conservative 1-hour eclipse duration per orbit, this creates an eclipse energy requirement of 207Wh. Assuming Li-ion battery efficiency of 95% [15], the requirement rises to 218Wh.

The high eclipse cycles of ~18000 cycles dictate the need for low Depth of Discharge (DoD) in the battery design. A DoD of 20% was chosen for this mission. This allows us to calculate the End of Life (EoL) capacity requirement for the battery, which was found to be 1090Wh. To minimize cell aging, the battery capacity at the beginning of life (BoL) should be 1635Wh, assuming 80% retained capacity at EoL. Also, a 20% margin is applied as a precautionary ECSS measure [16].

A 28-32V bus voltage is chosen to minimize resistive power losses associated with lower bus voltages. This aligns with the EPS hardware options for MOWI. Using 3.6V Li-ion cells, an 8S configuration gives 28.8V as desired. Table 3 below shows comparison between COTS vs space tech components for batteries.

| | Nominal Volatge (V) | Cell Capacity (Ah) | Cell Energy (Wh) | Configuration | Mass per config (kg) |
|---|---|---|---|---|---|
| 18650 Li-ion (COTS) [17] | 3.6 | 2.8 | 10.1 | 8S20P (1616Wh) | 7.68 |
| 21700 Li-ion (COTS) [17] | 3.6 | 4.5 | 16.2 | 8S13P (1685 Wh) | 7.28 |
| NCA (Space grade) [18] | 3.6 | 7.5 | 27 | 8S8P (1728 Wh) | 20.5 |

*Table 3: COTS vs Space Grade cell analysis. The cell energy is calculated from multiplying nominal voltage by cell capacity. The mass per configuation is derived from mass per cell.*

The energy per 8S string is 80.64Wh for the 18650 cells [17] and 129.6Wh for the 21700 [17] cells. To satisfy the baseline BoL capacity requirement of 1590Wh, an 8S13P configuration of 21700 cells is selected, providing 1685Wh of total capacity, which is greater than the required 1635Wh. Furthermore, this also offers the least weight penalty. The space-grade Lithium Nickel Cobalt Aluminum Oxide (NCA) cell [18] offers higher energy per cell (at 0.2C) at 27Wh, compared to 16.2Wh for 21700 cell. However, the NCA cell suffers from lower energy density per unit mass at 84 Wh/kg, compared to 242 Wh/kg for the 21700 cells. Thus, supporting the 21700 battery as a more optimal choice for this mission.

Note that heaters are required to keep the batteries operating at optimal temperatures of -40 to 60°C for discharge cycles and 0 to 60°C for charging cycles [17].

### 4.2.3 Solar Array Sizing

Solar irradiance at ~1AU is reported to be about 1361.6 ± 0.3 W/m², measured at the 2019 solar minimum [19]. For the baseline solar budget, the total energy that the arrays must provide per orbit equals sunlit load energy and the recharge energy from the eclipse.

$$E_{solar} = E_{sun} + \frac{E_{eclipse}}{0.9}$$

The 0.9 factor comes from the assumed Li-ion charging efficiency [15]. The sunlit load can be assumed to comprise of the different power draws from all systems except the heater. So $E_{sun} = 202Wh$ and $E_{eclipse} = 207Wh$ from before. Accounting for another 20% margin on the design, $E_{solar} = 519Wh$ or 519W of power needed per orbit. Below is a list of COTS vs space tech comparisons for solar arrays.

| | **Rocket Lab ZTJ+ (Space Grade)** [20] | **DHV array** [21] **using Azur 3G30A cells (COTS)** [22] |
|---|---|---|
| BoL Efficiency | 29.4 % | 29.0 % |
| BoL Power Density | 397.8 W/m$^2$ | 396.4 W/m$^2$ |
| EoL Power Density | 346.1 W/m$^2$ | 352.7 W/m$^2$ |
| Array area at EoL | 519/346.1 = 1.50 m$^2$ | 519/352.7 = 1.472 m$^2$ |
| Power remaining factor | Pmp = 0.87 | 25.8/29.0 = 0.8897 |
| Required Solar power at BoL | 519/0.87 = **597 W** | 519/0.8897 = **583.3 W** |

*Table 4*

Both the space grade solar array and the COTS option can provide 519W at EoL. Note that we are assuming ideal working conditions, such as normal sunlight incidence and no losses due to varying temperatures, contamination, degradation margins and Maximum Power Point Tracking (MPPT) efficiency.

The COTS DHV DSA-676W array is selected as a baseline option, as it is readily available deployable system rated at 676W BoL, exceeding the required 519W EPS power with adequate design margin. This simplifies implementation and reduces integration risk compared to custom solution. However, Rocket lab's ZTJ+ offers modular sizing, allowing choosing an array area to be more close to the required power needs. This optimization could result in weight savings and potentially lower the cost. Given that DSA-676W already satisfies the power budget and is readily implementable, it represents a more practical solution for this mission.

A Power Conditioning and Distribution Unit (PCDU) is now needed which accepts solar array input, perform MPPT, manage the battery, regulate the platform buses and distribute power to the subsystems simultaneously. Most high profile and large satellites use custom PCDU solutions, a COTS alternative offers lower cost, modularity and ease of implementation. An example of such is AAC Clyde Space's **STARBUCK-MINI** PCDU []. It can deliver up to 2.5kW of system power, regulate the solar array's MPPT and supplies a primary bus voltage of 28V as required. This hardware is also certified for radiation tolerance (TiD) = 20kRAD.

### 4.2.4 Component Resilience and Fault-tolerance

The components chosen may be suitable for the technical aspects of mission. However, they need to be assessed on their resilience and fault mitigation techniques when working in the lunar environment. Table 5 summarizes key resilience features and qualification testing levels.

The selected 21700 cells provide the desired battery capacity for this mission. On-board heaters will maintain the cells at their optimal working temperatures, enabling them to operate in the lunar thermal environment. However, they provide no radiation or acceptance testing and therefore rely on systems level mitigation. Resilience can be achieved through per cell monitoring, parallel string configuration and limiting DoD. This ensures limited degradation of the cells, extending their life cycle to meet the mission requirement.

The PCDU is radiation tolerant up to TRL-9 [23], which means its radiation tolerance is proven in actual missions. TVAC certification ensures that the component is suitable for the deep vacuum of space. The PCDU acts as the primary fault mitigation controller with its internal features. It is also certified for multi-year missions such as this one. Single Event Latchups are detected by the OBC and prompts the PCDU to cycle through the power system.

The solar array has undergone radiation testing according to ECSS compliance [24]. The use of integrated bypass diodes and MPPT limits damage from power surge when coming out of an eclipse. The honeycomb structure ensures mechanical stability during launch and thermal cycling. The coated coverglass provides some radiation shielding limits thermal expansion. Further electrical protection is achieved through PCDU's over-voltage, over-current and short-circuit protection.

Overall, subsystem level resilience is achieved through multiple layered protections incorporated by the components: radiation hardened solar array and PCDU, conservative energy storage system, actively monitored and controlled power conditioning. This ensures

slow degradation of the components in the lunar environment, meeting the long mission lifetime of 4 years.

| | **Radiation Tolerance** | **Thermal Range** | **Redundancy** | **Qualification Level** | **Other Safety features** |
|---|---|---|---|---|---|
| **21700 Cells** | - | -40°C - +60°C | Achieved though parallel strings and per cell monitoring | - | - |
| **DHV DSA-676W /** Azur 3G30A cells | Tested up to 1 MeV electron fluence levels: | 28°C – 140°C | | TRL8-TRL9, ECSS-E-ST-20-08C tested | Integrated Bypass Diode, Honeycomb core, 100 µm AR coated coverglass |
| **AAC Clyde Space Starbuck-Mini PCDU** | TiD of 20 kRAD (qualified >30 kRAD Si) | -30°C to +60°C | Redundant control buses | TVAC, TRL-9 | Protection on power lines(over-current, short circuit, over-voltage), fault-tolerant design |

*Table 5: A summary of features and specifications relevant to the product's space environment resilience.*

## 4.3 On-Board-Computing

### 4.3.1 Image analysis, resampling and compression

The MOWI imaging payload is sized to provide full coverage of both lunar poles to 60° latitude over the course of 4 years. To achieve this, each orbit will produce scan data covering a rectangle of approximately 100 m x 1820 km at each pole, given an average lunar radius of 1738 km. This data is provided as 8 x 6.59 GB LIDAR.

The raw data is far too voluminous to transmit back to Earth, so the raw LIDAR images will be processed on board the satellite with each full orbit being processed before the start of the next. Topographical maps of water concentration will be produced with a resolution constrained by the Lunar Pathfinder's data relay capabilities and MOWI's contact time with Pathfinder.

To minimise data storage and transmission requirements, topographical map data will be stored in the following structure:

- 1 image per orbital band per pole.
- 2 x 64-bit doubles representing the lunar latitude and longitude of the "top, left" corner of the map. "Top" is defined in the direction of MOWI's orbital track while "Left" is defined as looking directly down towards the Lunar surface perpendicular to MOWI's orbit track.
- A grid where each cell contains 2 numbers:
    - 1 x 16-bit signed integer elevation in metres, measured from the mean radius of 1737.4 km from the Moon's centre of mass. This is chosen noting that the maximum lunar surface elevation delta is under 20km.
    - 1 x 12-bit unsigned integer concentration water in percentage parts per million.

Given the constrained data rates and required error correction for relaying via Pathfinder, the expected worst case total data volume we can transmit per MOWI orbit on the payload downlink is 61 MBytes, with the best case being around 79 MBytes. Working backwards from this, we can calculate the best resolution map we can guarantee transmitting within this constraint.

$$\frac{511705088 - 128}{16 + 12} \times \frac{1}{2} = 9137588\ cells$$

The total area covered by each polar map band is $182 \times 10^6$ m$^2$ and therefore in our ideal case, each cell would represent a 4.46m x 4.46m square on the lunar surface. In the interests of ease of use we will resample the raw LIDAR data to a resolution of 5x5m and transmit that. That also gives us some additional safety margin on transmission.

To give us extra Payload Link capacity, each map file will be compressed using the standard ZIP algorithm as it is expected that there will be many recurring '0's in the data. The likely compression ratio is unknown and will require further analysis, however it is assumed that this will provide sufficient extra capacity to enable responding to TT&C data requests.

### 4.3.2 Microprocessor

An analysis of previous Lunar nanosat missions (See

Appendix C: Lunar Missions COTS analysis) revealed two TRL9 level COTS contenders for the OBC with lunar flight heritage:

- Space Micron Proton 400K Lite
- Sphinx C&DH (LEON 3 FT)

Of these, the Sphinx was chosen for MOWI because it provides a lot more design flexibility due it's combined custom CPU (LEON 3 [25], also used on other lunar missions in its own right [5]) and FPGA as well as having more recent heritage.

The Sphinx C&DH board, originally developed by JPL, has since been sold and re-sold and while the original is not available commercially, its successor is now available as a COTS product under the name "Gen7" from FrontGrade [26].

Some evidence exists [27] to suggest that the following radiation testing has been completed:

- TID: 100 krads (Si)
- TID Processor: GR712 Processor: 300 krads (Si)
- SEL: > 75 MeV cm²/mg or < $10^{-4}$ events per year
- LETth > 75 MeV cm²/mg $10^{-10}$ per bit per day or through application analysis
- LETth > 75 MeV cm²/mg $10^{-10}$ per bit per day or through application analysis

### 4.3.3 Storage and terrestrial SSD usage analysis

It is difficult to ascertain whether there are TRL9 SSDs available as COTS products, however some evidence suggests that the Mercury RH304T SSD [28], Exascend PR4 [29] and Phison 8TB SSD [30] have at least TRL6 for lunar orbit. Given flight-proven heritage is hard to find, the use of a non-space-qualified SSD is also assessed.

The primary concern for using non-space-qualified SSDs in space is radiation.

Ionising radiation concerns can be mitigated with 100mil Aluminium shielding added as a case to the SSDs.

Single Event Upsets however cannot be avoided altogether and would need to be handled using error detection and correction within the SSD. Standard COTS SSD datasheets do not include implementation details of their EDAC systems, so it is hard to do a paper analysis of a specific drive, however modern common implementation choices are well known. The default built in error correction in typical COTS SSDs is Low-Density-Parity-Checks (LDPC) and Bose-Chaudhuri-Hocquenghem (BCH) codes. LDPC has been determined to be very effective at correcting single-bit-errors [31], handling the expected error rate in lunar orbit ($10^{-6}$ bit-errors per day) while only requiring a scrub every roughly 32 days.

The challenge remains however that the control circuitry within non-space COTS SSDs is not radiation hardened and therefore invalidates their use in the space environment since the controller will likely cause operational errors or suffer permanent failure. While this could to some extent be mitigated by running two commercial SSDs in a RAID1 configuration with potentially some additional circuitry to handle unexpected operational errors, the likelihood of

failure is still very high compared to a controller that has been built radiation hardened from the outset.

In comparison, the Mercury RH304T SSD has been designed for the space environment from the outset. The controller is radiation hardened and has been tested above radiation levels expected in interstellar space and a custom error correction protocol has been implemented on the NAND array to keep bit-errors low. The SSD also mitigates against ionising radiation to well above our requirements in lunar orbit so no additional shielding would be needed.

- Total ionizing dose (TID) > 100 krad (all components except NAND)
- TLC NAND in SLC mode. SEE/TID testing in progress; 30 krad TID expected
- Configuration upsets immunity to LET > 103 MeV.cm²/mg
- Single-event latch-up (SEL) immunity to LET > 103 MeV.cm²/mg
- Registers SEU rate < $10^{-12}$ errors/bit-day (GEO solar min)
- Single-event transient (SET) upset rate < $10^{-8}$ errors/bit-day (GEO solar min)
- Total Ionising Dose (TID) > 100 krad

Therefore, a single Mercury RH304T SSD will be used to provide 4.5TB of storage for raw LIDAR images waiting to be processed as well as analysed water maps and historical telemetry data.

An additional external SpaceVPX SRIO bridge module would be needed to interface the SSD with the OBC via, for example, SPI.

### 4.3.4 Interfaces

The following interfaces with the OBC are available for use on the satellite and are estimated to be sufficient for the devices to be connected to the OBC.

- 2x SpaceWire interfaces with RMAP (GR712RC)
- Up to 8x Asynchronous RS422 serial links with added Mezzanine Card
- 4x SPI Master Ports
- Single I2C multi-master core, with 7-bit or 10-bit mode slave-select addressing (GR712RC)
- 32 GPIO
- 2x JTAG (processor and FPGA)
- 1 Ethernet PHY RMIII (for GSE)
- 26 Analog channels with added Mezzanine Card
- Interfaces with IRIS and ACS systems (Star Tracker, IMU, Sunsors, Cold Gas)

Their usage is outlined in more detail in 2.2 Internal Architecture.

# 5 References


[1] C. Bridges, “EEEM059 Space Avionics Assignment 1: Platform Avionics Design,” 2025.

[2] R. Providence, “Researchers create first global map of water in Moon’s soil,” Brown University, 13 September 2019. [Online]. Available: https://www.brown.edu/news/2017-09-13/moonwater. [Accessed 27 February 2026].

[3] SSTL, “Lunar Pathfinder: Data relay satellite in orbit around the Moon - SERVICE USER GUIDE - V003,” 2020.

[4] S. R. Sadin, F. P. Povinelli and R. Rosen, “The NASA Technology Push towards Future Space Mission Systems,” *Acta Astronautica,* vol. 20, pp. 73-77, 1989.

[5] N. Baresi, H. J. Holt, N. Bernardini, X. Fu, M. Tisaev, Y. Gao, C. Bridges, A. Lucca Fabris, R. P. Armellin and P. Murzionak, “Mission Analysis and Design of VMMO: The Volatile Mineralogy Mapping Orbiter.,” 2021.

[6] M. D. Looper, J. E. Mazur, J. B. Blake, H. E. Spence, N. A. Schwadron, J. K. Wilson, A. P. Jordan, C. Zeitlin, A. W. Case, J. C. Kasper, L. W. Townsend and T. J. Stubbs, “Long-Term Observations of Galactic Cosmic Ray LET Spectra in Lunar Orbit by LRO/CRaTER,” *Space Weather,* vol. 18, no. 12, 2020.

[7] J. E. Mazur, W. R. Crain, M. D. Looper, D. J. Mabry, J. B. Blake, A. W. Case, M. J. Golightly, J. C. Kasper and H. E. Spence, “New measurements of total ionizing dose in the lunar environment,” *Space Weather,* vol. 9, no. 7, 2011.

[8] M. Harrell, G. Schroeder and S. Daire, “Lunar Environment, Overview,” in *Handbook of Life Support Systems for Spacecraft and Extraterrestrial Habitats*, Springer International Publishing, 2021.

[9] “SatBase - Largest Spacecraft Marketplace,” SatBase, [Online]. Available: https://www.satcatalog.com/component/sbtrcvr-s-band-transceiver/.

[10] “IQ Technologies for Earth and Space GmbH,” Satbase, 07 2025. [Online]. Available: https://www.iq-spacecom.com/images/Datasheets/IQspacecom_S_Band_Antenna_datasheet_07.2025.pdf?utm_source=copilot.com.

[11] P. Z. S. Dr. Atta Quddus, *EEEM031 - Fundamentals of Satellite Communications, Satcom Assignment 2025/2026, "Internet via "UltraSat" Ka-Band Satellite",* University of Surrey, 2022.

[12] CCSDS, CCSDS RECOMMENDATIONS FOR RADIO FREQUENCY AND MODULATION SYSTEMS, Washington, DC, USA: CCSDS Secretariat , 2021.

[13] A. O’Dea, “206 Telemtery General information,” 2023. [Online]. Available: https://deepspace.jpl.nasa.gov/dsndocs/810-005/.

[14] E. C. f. S. Standardization, “Adoption Notice of CCSDS 131.0-B-3, TM Synchronization and Channel Coding,” 01 March 2021. [Online]. Available: https://ecss.nl/wp-content/uploads/2021/03/ECSS-E-AS-50-21C(1March2021).pdf.

[15] K. L. King Jet Tseng, “Energy efficiency of lithium-ion battery used as energy storage devices in micro-grid,” *IEEE Xplore,* 2016.

[16] ESA-ESTEC, “ECSS-E-ST-20-01C – Multipactor design and test,” *ECSS,* 2020.

[17] “Lithium-Ion Batteries (LIB),” Mitsubishi Electric, 2026. [Online]. Available: https://www.mitsubishielectric.com/. [Accessed March 2026].

[18] E. B. GmbH, “UHP341440 NCA,” EAS Batterie, [Online]. Available: www.eas-batteries.com. [Accessed March 2026].

[19] N. A. K. J. D. H. Sami K. Solanki, “Solar Irradiance Variability and Climate,” *Annual Review of Astronomy and Astrophysics 323(1),* 2013.

[20] S. T. C. Rocket Lab, “ZTJ+ Space Solar Cell,” 2023. [Online]. Available: https://rocketlabcorp.com/assets/RL-SolAero-Data-Sheet-ZTJ+-v7.pdf. [Accessed 1 March 2026].

[21] D. Technology, “COTS Solar Arrays for Satellites,” DHV Technology, 2024. [Online]. Available: https://dhvtechnology.com/products/cots-solar-arrays-satellites/. [Accessed March 2026].

[22] AZUR SPACE Solar Power, “30% Triple-Junction (TJ) III-V Bare Solar Cells & Assemblies,” AZUR SPACE Solar Power, 11 April 2025. [Online]. Available: https://www.azurspace.com/en/. [Accessed March 2026].

[23] U. R. a. Innovation, “Eligibility of technology readiness levels (TRL),” 2026. [Online]. Available: https://www.ukri.org/councils/stfc/guidance-for-applicants/check-if-youre-eligible-for-funding/eligibility-of-technology-readiness-levels-trl/. [Accessed March 2026].

[24] E. C. f. S. Standardization, “ECSS-E-ST-20-08C Rev.1 – Photovoltaic assemblies and components (18 July 2012),” 18 July 2012. [Online]. Available: https://ecss.nl/standard/ecss-e-st-20-08c-rev-1-photovoltaic-assemblies-and-components-18-july-2012/. [Accessed March 2026].

[25] ESA, “ESA - Microprocessors,” [Online]. Available: https://www.esa.int/Enabling_Support/Space_Engineering_Technology/Onboard_Computers_and_Data_Handling/Microprocessors. [Accessed 24 February 2026].

[26] FrontGrade, “Generation 7 Single Board Computer (Gen 7),” [Online]. Available: https://www.frontgrade.com/Gen7-SBC. [Accessed 25 02 2026].

[27] K. F. Ortega and M. Roche, “Sphinx CDH and the open-source F Prime Sphinx Reference Deployment,” JPL, 8 February 2021. [Online]. Available: https://youtu.be/oUn-Q1ro-vI?si=mjStx0-O8ClZLJPw. [Accessed 1 March 2026].

[28] Mercury, “RH304T Solid-State Data Recorder,” [Online]. Available: https://www.mrcy.com/application/files/6716/3250/7093/5094.01E_RH304T_3U_VPX_SRIO_SSDR.pdf. [Accessed 29 February 2026].

[29] Nex-Pro Global, “PR4 Series,” [Online]. Available: https://nexpro-global.com/pr4-series/. [Accessed 28 February 2026].

[30] Phison, “Phison’s 8TB SSD Ready for Historic Liftoff After Earning NASA Certification,” 6 December 2022. [Online]. Available: https://www.phison.com/en/category/article/press-releases/phisons-8tb-ssd-ready-for-historic-liftoff-after-earning-nasa-certification. [Accessed 28 February 2026].

[31] S. Jeon, B. V. K. Vijaya Kumar, E. Hwang and M. K. Cheng, “Evaluation of Error-Correcting Codes for Radiation-Tolerant Memory,” *IPN Progress Report,* pp. 42-181, 15 May 2010.

[32] NASA, “Appendix S Concept of Operations Annotated Outline,” 21 February 2026. [Online]. Available: https://www.nasa.gov/reference/appendix-s-concept-of-operations-annotated-outline/.

[33] K. L. Ryder and M. J. Campola, “The Lunar Radiation Environment,” 4 April 2022. [Online]. Available: https://ntrs.nasa.gov/api/citations/20220007070/downloads/20220007070-Kaitlyn-Ryder-2022-LaRC-Presentation_v2.pdf. [Accessed 27 February 2026].

[34] C. Bridges, “2025 SA-05.1,” University of Surrey, 2025.

# 6 Appendices

## 6.1 Appendix A: Orbital Parameters

The basic orbital parameters of both MOWI and the Lunar Pathfinder are listed here.

| | **Lunar Pathfinder** | **MOWI** |
|---|---|---|
| *Orbit Type* | | Circular LLO |
| *Altitude* | Pe: 500km<br>Ap: 7500 km | 100 km |
| *Semi-major axis* | 5737.4 km | 1837.4 km |
| *Eccentricity* | 0.61 | 0 |
| *Inclination* | 57.82° | 90° (polar) |
| *Argument of Pericentre* | 90° | N/A |

*Table 6 - Orbital Parameters for MOWI and Lunar Pathfinder*

## 6.2 Appendix B: Acronyms

- MOWI – Moon Orbiter for Water Imaging
- TRL – Technology Readiness Level
- TT&C – Tracking, Telemetry and Command
- OBC – On Board Computer
- COTS – Commercial Off-The-Shelf
- FPGA – Field Programmable Gate Array
- AOCS – Attitude and Orbit Control Systems
- C&DH – Command and Data Handling
- RAID – Redundant Array of Independent Disks
- SPE – Solar Particle Event
- GCR – Galactic Cosmic Ray

## 6.3 Appendix C: Lunar Missions COTS analysis

Analysis conducted of https://www.nanosats.eu to determine which COTS products were used in all known lunar nanosat missions.

### 6.3.1 OBC, AOCS and Solar Panels

| Mission | OBC | AOCS | Solar Panels |
|---|---|---|---|
| BioSentinel | | Blue Canyon XACT | |
| ArgoMoon | | Blue Canyon XACT | DHV |
| LunaH-Map | | | MMA Design eHaWK |
| Lunar IceCube | Space Micro Proton 400K Lite | Blue Canyon XACT | Pumpkin |
| EQUULEUS | | Blue Canyon XACT-50 | |
| OMOTENASHI | | Blue Canyon XACT | |
| Lunar Flashlight | Sphinx C&DH (LEON 3 FT) | Blue Canyon XACT | Blue Canyon, MMA Design Trifold Solar Panels |

### 6.3.2 Thruster, Avionics, Transceiver, EPS and Platform

| Mission | Thruster | Avionics | Transceiver | EPS | Platform |
|---|---|---|---|---|---|
| CAPSTONE | Stellar Exploration | | | | |
| LunaH-Map | Busek BIT-3 | Blue Canyon XB1 | JPL Iris Deep Space Transponder | | |
| Lunar IceCube | Busek BIT-3 | | | Pumpkin | |
| EQUULEUS | Pale Blue | | | | |
| LunIR | | | | | Tyvak |
| Lunar Flashlight | Bradford ECAPS | | JPL Iris Deep Space Transponder | | |

## 6.4 Appendix D: Communication Sub-system

### 6.4.1 Formulae Used

- Orbital Period, $T = 2\pi\sqrt{\frac{a^3}{\mu}}$
- Visibility half angle, $\theta = \cos^{-1}(\frac{R_{Moon}}{r})$
- Orbital angular rate, $\omega = \frac{360^0}{T}$
- $EIRP = PTx + G_{antenna} + L_{cable}$
- $FSPL = 20\log_{10}(\frac{4\pi d}{\lambda})$
- $C/No = EIRP - FSPL - L_{OtherLosses} + G/T + 228.6$
- $Eb/No = C/No - 10\log_{10}(R_b)$
- $Power_{PA} = \frac{P_{RF}}{\eta}$

### 6.4.2 Calculations

#### 6.4.2.1 Relative range and time

**Range**

Minimum possible range: When the LP is at periselene (~673 km) and the MOWI is at 100 km

d_min = 673-100 = 573 km

Geometery offset, d_min ~ 600 km

Maximum possible range: When the LP is at aposelene (~7331 km) and the MOWI is at 100 km.

d_max = 7331+100 = 8331 km

Geomtery offset, d_max ~ 8000-9000 km

**Time**

- Visibility half angle, $\theta = \cos^{-1}(\frac{R_{Moon}}{r})$

  For periselene,
  $\theta = 43.8$

  total visibility arc = 2 x 43.8 ~ 88 degrees

Orbital angular rate, $\omega = \frac{360^0}{T}$=3.07 deg / min

Visibility time = 88/3.07 ~ 29 minutes
For aposelene,
Visibility time ~ 51.8 minutes
**Average time per orbit ~ 40-50 minutes**

#### 6.4.2.2 Data Volume-Reasoning

According to the requirements of the mission, the storage of around 8x6.48 GB per orbit has been asked to be maintained. Taking that into consideration, the maximum data supposed to be generated per orbit is 8 images x 6.48 GB per orbit which is nearly equal to 51.84 GB and 414.72 Gbits per orbit which is massive to collect, store and send. The limits given for communication to perform TT&C are 48 kbps BPSK and 1 Mbps QPSK. The S-band transceiver selected supports simultaneous transmitting and receiving chains, that enables full-duplex as mentioned in the mission requirements. The data generated vastly exceeds the link capacity. For example, at 1 Mbps, which is the upper limit of the communication, it would nearly take 115 hours to downlink that data, which is clearly infeasible. Even at the maximum capacity of the Lunar Pathfinder which is 2 Mbps, it would approximately take 58 hours. The Lunar Pathfinder CubeSat user example also expects only ~500 Mb/day [3]. There is massive difference between the data asked to generate and data able to send. Therefore, to make this mission work with the mentioned requirements, a solution to the problem of data generation has been given in the OBC Subsystem section further in the report. From the Communication point of view, calculation needs to be done to provide OBC Subsystem with the maximum usable data per pass.

**Required Downlink Calculation:**

The required downlink per orbit can be determined by considering the minimum and maximum visibility window, which is 40 minutes and 50 minutes respectively. In the 40 mminute window, the following data rates can be expected:

- 1 Mbps for 20 minutes
- 256kbps for 10 minutes
- 48.8kbps for 10 minutes

| 1 mbps transmission | |
|---|---|
| Raw Rate | 1000000 bits/s |
| LDPC 1/2 | 500000 bits/s |
| Protocol Overhead | 10% (90% usable) |

For 1 Mbps,

Usable bits after the protocol head and LDPC ½ coding = 450 kbps

Close approach ~ 20 min ~ 1 Mbps

Therefore, 20x60 = 1200s

**Data per pass = 450,000 x 1200 ~ 67.5 MB per pass**

Similarly for 256 kbps ~ 12 minutes

**Data per pass = 10368 MB**

and for 48.8 kbps ~ 8 minutes

**Data per pass = 1.31 MB**

**Total usable data per pass = 79.178 MB per contact orbit**

Assumed favourable contact orbits out of 12.2 orbits = 2.25 ~ 3

Therefore, **Total usable data per day ~ 178.1-237 MB per Earth day.**

The following link budget calculation has been done by using the formula mentioned in **5.4.1**

| | |
|---|---|
| EIRP | 17 dBW |
| FSPL | $FSPL_{600}$ ~155 dB |
| | $FSPL_{3000}$ ~ 169 dB |
| | $FSPL_{8000}$ ~ 177.5 dB |
| C/No | $C/No_{600}$ ~ 78.4 dBHz |
| | $C/No_{3000}$ ~ 64.5 dBHz |
| | $C/No_{8000}$ ~ 56.5 dBHz |
| Required EIRP | 16.53 dBW |
| Required Antenna Gain | 10.53 dBi |
| Required TX Power | ~4.5 W |

| | 48.8 kbps | 256 kbps | 1 Mbps | Mode/Range |
|---|---|---|---|---|
| Eb/No | 1. 31.6 dB<br>2. 17.6 dB<br>3. 9.6 dB | 1. 24.4 dB<br>2. 10.4 dB<br>3. 2.4 dB | 1. 18.5 dB<br>2. 4.5 dB<br>3. -3.5 dB | 1. 600 km<br>2. 3000 km<br>3. 8000 km |
| Required Eb/No | ~2 dB | | | |
| Link Margin | 1. +29.6 dB<br>2. +15.6 dB<br>3. +7.6 dB | 1. +22.4 dB<br>2. +8.4 dB<br>3. +0.4 dB | 1. +16.5 dB<br>2. +2.5 dB<br>3. -5.5 dB | |

## 6.5 Appendix F: Requirements

### 6.5.1 Mission Requirements

| Req # | Description | Parent | Reference | Change Log | Assumptions | Constraints |
|---|---|---|---|---|---|---|
| | | | | | | |
| **MIS-1** | MOWI shall maintain a 100km circular polar orbit (LLO) around the Moon with a nodal precession of 0.1232° per day | OB-1, MIS-3 | | [28/02/2026] Agreed circular orbit with an inclination of 90° to simplify the mission<br>[01/03/2026] Added nodal precession requirement to ensure full coverage of the polar region in the mission time frame | Assuming eccentricity is perfectly 0 and standard lunar orbital perturbations dealt with autonomously with electric propulsion | - |
| **MIS-2** | MOWI shall scan for water at both lunar poles on every orbit above 60° latitude | OB-1 | - Both poles: Bridges, C.<br>- 60°: [Pro19] | [24/02/2026] Clarified requirement of both poles<br>[27/02/2026] Updated to 60° latitude | Assuming sufficient LIDAR coverage | - |
| **MIS-3** | MOWI shall complete a full detailed scan of both poles within 4 years | OB-3 | | | Constant high resolution Lidar data captured. Nodal precession allowing full coverage of the poles | Missed payload opportunities due to anomalies. |
| **MIS-4** | MOWI shall maintain continuous operations for a minimum of 4 years | MIS-3 | | | Components are sized according to EoL specs. | Long-term degradation of spacecraft systems/components. Unexpected failures limiting lifetime. |
| **MIS-5** | MOWI shall transmit the last orbit's scan data on initiation of Payload Link by the Lunar Pathfinder | OB-1 | | [26/02/2026] Changed from last month to last orbit only<br>[27/02/2026] Changed to initiation by Lunar Pathfinder following agreement on using "Autonomous" SLA<br>[28/02/2026] Agreed on "Payload Link" as name of data downlink to Pathfinder | Lunar Pathfinder's autonomous service initiating the link successfully. Last orbit's scan data is compressed/processed before transmission | Contact time per day and service allocation |
| **MIS-6** | MOWI shall commence streaming of current telemetry data over TT&C Link on | MIS-4 | | [01/03/2026] Streaming of telemetry added | TT&C link available even at the largest separation. | 48kbps TT&C link speed |

| | | | | | | |
|---|---|---|---|---|---|---|
| | initiation of TT&C Link by the Lunar Pathfinder | | | | | |
| **MIS-7** | MOWI shall transmit historical map data on request via the Payload Link | MIS-3 | | [01/03/2026] Agreed to respond on Payload Link rather than TT&C Link | On-board storage retains past payload data for a certain defined period. Also assuming bandwidth headroom to accommodate extra data | Storage limits amount of past data that can be stored. Limited bandwidth for extra data per transmission |
| **MIS-8** | MOWI shall transmit historical telemetry data on request via the Payload Link | MIS-4 | | [01/03/2026] Agreed to respond on Payload Link rather than TT&C Link | Date and time data are properly synchronized | Limited bandwidth for extra data per transmission |

### 6.5.2 System requirements

| Req # | Description | Parent | Reference | Change Log | Assumptions | Constraints |
|---|---|---|---|---|---|---|
| | | | | | | |
| **SYS-1** | Mission data shall consist of a topographical water density map data | OB-1 | | | Raw data is processed on-board to map water density. Map coordinate and reference frame included in the mission data | Map resolution limited by downlink volume and on-board processing time |
| **SYS-2** | The system should maintain a minimum safety margin of 1.5 | MIS-4 | | | Margin applied at subsystem level | Certain data like daily contact period cannot be margined |
| **SYS-3** | The system will maintain sufficient telemetry and engineering metrics to enable visibility of current operational status and provide sufficient data for diagnostics | MIS-4 | | | Telemetry includes additional health data for EPS, Comms, propulsion and payload systems. Crucial health data is streamed in real-time and not just stored for later downlink | Sample rate for crucial health and operational data |
| **SYS-4** | The spacecraft should have redundant systems in place | SYS-2 | | | Achieved though component-level redundancy and/or subsystem level processing | Mission budget and constraints restrict full hardware redundancy |
| **SYS-6** | MOWI shall need to endure ~17,900 eclipse cycles over the mission lifetime | MIS-4 | | | Assuming ~12.2 orbits per day for 4 years and one eclipse per orbit | Components limited lifetime, lunar environment degradation. |
| **SYS-7** | There will be 3 operating modes: Nominal, Payload Collection and Data Downlink | | [Bri25] | | Spacecraft operations are well defined for each mode | MOWI must remain energy-positive for each mode |
| **SYS-8** | Operating modes must be able to run concurrently | SYS-7 | | [01/03/2026] Added to ensure we can handle comms with Pathfinder at any point in the orbit | Spacecraft operations can overlap between modes | Peak power limitation may force partial concurrency |
| **SYS-9** | System must accommodate temperature swings of +127°C to -183°C | MIS-1 | [Har21] | | Assuming each component is designed or has systems in place to manage temperature | Fatigue from high frequency and long period thermal cycling |
| **SYS-10** | System must handle SEUs caused by Solar particles and GCRs at a rate of 1E-6 on all bits per day | MIS-1 | [Bri251] | [02/03/2026] Value refined following additional research | EDAC implemented for critical data and storage and safe recovery mode exists | It is assumed that this measure applies to certain densities of transistor arrays but it is unknown what that density is |

| | | | | | | |
|---|---|---|---|---|---|---|
| **SYS-11** | MOWI shall use COTS products where possible | OB-2 | | | Assuming the COTS selection is suitable for the mission and the lunar environment | Some COTS may not be tested in space |
| **SYS-12** | All components must operate effectively considering ionising radiation of 5E-7 rads/sec in Silicon = 63 rads over the 4 years mission duration | MIS-1 | [Maz11] | [28/02/2026] Value refined following additional research | 100mil of Aluminium shielding is applied in all cases to achieve this dosage | Shielding can be limiting and not applicable to all parts of the spacecraft |

### 6.5.3 Communication Subsystem Requirements

| Req # | Description | Parent | Ref | Change Log | Margin | Assumptions | Constraints |
|---|---|---|---|---|---|---|---|
| | | | | | | | |
| **COM-1** | Comms shall support 1mbps S-band QPSK downlink to the Lunar Pathfinder | MIS -7 | 4.1.1 | [25/2/26] Defining mission operations | Positive link margin between separation of 600km and 300km | Assuming a constant 1mbps for 20mins | When link margin is negative >8000km |
| **COM-2** | Comms shall support 48kbps BPSK full-duplex TT&C at maximum separation | MIS-6 | 4.1.1 | [25/2/26] Defining mission operations | Positive link margin at largest separation (+7.6 dB) | Assuming TT&C link is available for full orbital period | Continuous power consumption |
| **COM-3** | Comms shall adaptively switch between 1mbps and 256kbps payload data rate depending on separation distance | OB1 | 4.1.2.3 | [1/3/26] Adding a third communication mode | 256kbps - 1mbps | Assumes 40min contact time per orbit | Not all orbits allow for contact |
| **COM-4** | Comms shall allow a downlink capacity of 178-237Megabytes per Earth day | OB1 | 4.1.2.3 | [1/3/26]Modifying the calculated downlink capacity | 178MB - 237MB | Assumes 90min contact time and 2.25-3 favourable orbits per day | Pathfinder contact time of 90min per day |
| **COM-5** | Comms shall require EIRP of 16.5dBW | COM-1 | Appendix E: 5.4.2.2 | [1/3/26] Calculated the required EIRP based on new design margin | ~16.5dBW | Assuming 1dB cable loss, 2dB system loss and 11dBi antenna | Power limit of 22W and PA of 35% |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **COM-6** | Comms transceiver shall function optimally in the lunar environment over the mission lifetime | SYS-12 | 1 | [2/3/26]Made COTS transceiver selection [25/2/26] Defining mission operations | COTS equipment rated for ~20krad and >3 years lifetime | Assuming lunar radiation dose of 63 rad over lifetime | >3 year lifetime |
| **COM-7** | Comms antenna shall function optimally in the lunar environment over mission lifetime | SYS-12 | 1 | [2/3/26]Made COTS antenna selection<br>[25/2/26] Defining mission operations | TRL-9 certified | Assuming compatibility with Lunar TiD and 4 years lifetime | Optimized for LEO orbits |

### 6.5.4 Power Subsystem Requirement

| Req Number | Description | Parent | Reference | Change Log | Margin | Assumptions |
|---|---|---|---|---|---|---|
| **EPS-1A** | The EPS shall supply 207W at peak load inc heater(5W) | MIS-4 | Section 4.2.2 | [1/3/25] Made changes to increase comms power from 16W to 22W [28/2/25] Changed the comms power to 16W | Additional 11W is deliverable by the EPS system | Assuming battery efficiency of 95%, requirement rises to 218W |
| **EPS-1B** | The EPS shall support 80W for electric propulsion at any given mode | SYS-8 | Section 4.2.1 | [25/2/25] Started designing the system | - | Needed for orbital perturbation corrections |
| **EPS-1C** | The EPS shall tolerate single point faults without loss of function | SYS-4 | Section 4.2.4 | [2/3/25] Evaluated EPS resilience | - | Assuming no critical damage sustained by components |
| **EPS-2A** | The battery is sized accordingly to meet the capacity at BoL and EoL | MIS-4 | | [1/3/25] Calculated final battery values [28/2/25] Determined initial values | 1635W at BoL and 1090Wh at EoL | Assuming 20% design margin and 80% retained capacity at EoL |
| **EPS-2B** | Battery DoD shall be limited to 20% to meet mission's ~18000 cycles | SYS-6 | | [25/2/25] Started designing the system | 20% | Assuming the cells do not degrade further due to other losses such as radiation damage. And modelling eclipse duration to be ~1h |
| **EPS-2C** | The battery shall be maintained within the designed thermal limits | SYS-9 | | [1/3/25] Analysed the battery thermal limits | −40 to +60°C for discharge and 0 to +60°C for charge cycle | Assuming the 5W heater is capable of maintaining the battery at these temperatures |

| | | | | | | |
|---|---|---|---|---|---|---|
| **EPS-3A** | The solar array shall provide operational power and recharge the cells under sunlit conditions | SYS-6 | | [25/2/25] Started designing the system | Needed power = 519W (EoL) | Assuming Li-ion charging efficiency of 0.95 |
| **EPS-3B** | The solar array shall be sufficient to meet EoL and BoL power requirements | MIS-4 | | [1/3/25] Analysed the solar power requirements | 519W at EoL and 597W at BoL | Assuming no losses due to contamination, radiation and varying temperatures |
| **EPS-4A** | The PCDU will utilize MPPT to maximize solar power over varying conditions | SYS-9 | | [1/3/25] Analysed the PCDU functions | +127℃ to -183℃ | Assuming the 4 independent MPPT can successfully maintain high power mode |
| **EPS-4B** | The PCDU will handle battery management, including charging, current limiting, over-voltage, short-circuit | SYS-7 | | [1/3/25] Analysed the PCDU functions | 28-32V Bus | - |
| **EPS-4C** | The PCDU shall supply regulated power to each of the subsystem | SYS-7 | | [25/2/25] Started designing the system | 28V for EP, 12V for Comms, 5V for OBC | - |
| **EPS-4D** | The PCDU shall handle electrical protection over the power buses including over current, over-voltage, short-circuit. | SYS-4 | | [1/3/25] Analysed the PCDU functions | - | - |

### 6.5.5 OBC Subsystem Requirements

| Req # | Description | Parent | Reference | Change Log | Assumptions | Constraints |
|---|---|---|---|---|---|---|
| | | | | | | |
| **OBC-1A** | OBC must handle housekeeping collection at 60 kbps per orbit in appropriate buffers. | SYS-3 | [Bri25] 4.3.3 | [27/02/2026] Split out to OBC-1a/b as these are handled separately | - This will comprise all digital and converted analog data from all sub-systems.<br>- This is the full data collection needed by Nominal Mode operation. | |
| **OBC-1B** | Storage for 8 x 6.48 GB LIDAR images per orbit is needed | MIS-2 | [Bri25] 4.3.3 | [27/02/2026] Split out to OBC-1a/b as these are handled separately | - The imaging payload scan width will be 100m<br>- The 8 x 6.48 GB raw data covers the combined 120 degree arc per pole | |
| **OBC-2** | OBC shall do onboard processing of LIDAR images to produce a topographical map of water content at the poles | SYS-1 | 4.3.1 | | - The data analysis software will be sufficiently optimised to process all raw data within an orbit (118 minutes) | - Speed of the chosen CPU<br>- data access rates on The SSD |
| **OBC-2-1** | The topographical map shall have a resolution of 5 metres per pixel | | 4.3.1 | [01/03/2026] Updated transmission capacity of 79 MB per orbit has enabled potential resolution improvement to 4 m per pixel, however lower bound of 61 MB is still being used to | - We are not limited by the resolution of the raw LIDAR data in this regard | Resolution limited by Payload Link transmission capabilities |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | mitigate risk of running out of capacity for responding to TT&C requests | | |
| **OBC-2-2** | Map data shall be compressed for storage and transmission to minimise data load | SYS-1 | 4.3.1 | | Data compression will provide a larger comms window per orbit in which TT&C data requests can be fulfilled | Resolution limited by Payload Link transmission capabilities |
| **OBC-3** | As much historical map data will be stored as feasibly possible for later retrieval by TT&C request | MIS-7 | 4.3.3 | | - Data will be deleted on a FIFO basis to maintain a rolling window<br>- Available storage will be segmented to safeguard both historical telemetry data as well as historical map data | - The total storage size we can cost-effectively include, balanced by the power budget.<br>- Storage is shared with Map data |
| **OBC-4** | As much historical telemetry data will be stored as feasibly possible for later retrieval by TT&C request | SYS-3, MIS-8 | | | - Data will be deleted on a FIFO basis to maintain a rolling window<br>- Available storage will be segmented to safeguard both historical telemetry data as well as historical map data | - The total storage size we can cost-effectively include, balanced by the power budget.<br>- Storage is shared with Map data |
| **OBC-5** | All OBC components must have sufficient radiation hardening | SYS-12 | 4.3 | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | and / or shielding to accommodate ionising radiation | | | | | |
| **OBC-6** | All OBC components must have sufficient resilience to thermal variation | SYS-9 | 4.3 | | | |
| **OBC-7** | The OBC must be "radiation tolerant" to SEUs at the required rate for lunar orbit | SYS-10 | 4.3 | | | |
| **OBC-8** | Long-term storage shall be shielded with 100mil Aluminium | SYS-12 | 4.3.3 | | | |
| **OBC-9** | Long-term storage shall for radiation-tolerant to SEUs at the required rate for lunar orbit | SYS-10 | 4.3.3 | | | |